

Desktop Metrics

Specify which services you'd like to monitor for the Windows machine.

☒ **Ping**
Monitors the machine with an ICMP "ping". Useful for watching network latency and general uptime.

☒ **CPU**
Monitors the CPU (processor usage) on the machine.
Warning Load: % Critical Usage: %

☒ **Memory Usage**
Monitors the memory usage on the machine.
Warning Usage: % Critical Usage: %

☒ **Uptime**
Monitors the uptime on the machine.

☒ **Disk Usage**
Monitors disk usage on the machine.
Drive: Warning Usage: % Critical Usage: %
Drive: Warning Usage: % Critical Usage: %

Figure 3: Desktop metrics settings

Monitoring Windows Server

Now, let's monitor the Windows Server, just as we have monitored the Windows desktop.

The steps to be followed for Windows Server are listed below:

1. First, the agent NSClient++ must be installed in our Windows machine.
2. Inside NagiosXI, we have to configure the Windows server option and then assign the IP address 192.168.5.67 (for example) to the Windows server.

Settings for the server metrics: Just as we did for the Windows desktop, we can set our own parameters for monitoring the servers. For CPU, memory usage, disk usage, etc, we can set our own threshold value, below which the 'soft state' will be converted into the 'hard critical' and 'hard warning' state. The server metrics is shown in Figure 4.

Server services: We can also monitor any services, processes as well as the performance counter of our services, if we want. And the results of this monitoring will be displayed in our dashboard. The services added are shown in Figure 6. We can monitor the SQL server as well as the IIS Web servers.

Performance counter: In the performance counter, we are monitoring three major parameters such as page file usage, log on error and server work queue.

We can manually specify the values for critical and warning states. For implementation, we have specified the warning states as: 70 pages for file usage, two for log on errors and a warning of four for the server work queue. This means that if our server processes more than four tasks in a queue then the state will change, and NagiosXI will generate the critical message to the admin to correct the error. The performance counter is shown in Figure 5.

Server Metrics

Specify which services you'd like to monitor for the Windows server.

☒ **Ping**
Monitor the server with an ICMP ping. Useful for watching network latency and

☒ **CPU**
Monitors the CPU (processor usage) on the server
 % %

☒ **Memory Usage**
Monitors the memory usage on the server.
 % %

☒ **Uptime**
Monitors the uptime on the server.

☒ **Disk Usage**
Monitors disk usage on the server.

Drive: % %

Drive: % %

Drive: % %

Drive: % %

Drive: % %

[Add Row](#) | [Delete Row](#)

Figure 4: Server metrics

Server monitoring settings: We can also determine the maximum number of checks for the desktop. NagiosXI will keep on checking till the count reaches the maximum count value. Once the value reaches its maximum, NagiosXI will notify the admin about the error with the help of alert messages and e-mails.

Status generation for Windows Server: The status graph of Windows Server contains all the readings of the various parameters that we have initially set. The two services, IIS Web service and SQL service, are both monitored. Both the services are also showing as unknown since the services are not started in the server. Our performance counters are also shown in the graph. The status of the log on errors, page file usage and the server work queue shows 'OK'. All the server metrics settings, such as CPU usage, Disk D and memory usage, also shows 'OK', but Disk C is shown as being in a 'critical state' as the attempt count has reached a maximum level.

Graph generation: The graph is made based on device

Performance Counters

Specify any performance counters that should be monitored.

Performance Counter	Display Name	Counter Output Format	Warning Value	Critical Value
☒ \Paging_File_Total\%Usage	Page File Usage	Page File Usage is %2.1%	70	80
☐ \Server\Errors\System	Login Errors	Login Errors since last reload	2	20
☐ \Server\Work_Queue\WorkQueue	Server Work Queue	Current work queue (an ind)	4	7
☐				
☐				
☐				

Figure 5: Performance counters